

scripts/run-vm-signing-matrix.sh

— User Guide

Last verified: 2026-05-16 (Phase 6a + 6b §11.4.29 snake_case migration: scripts/run-vm-signing-matrix.sh path-references updated from Submodules/ to submodules/) **Inheritance:** HelixConstitution §11.4.18 (script documentation mandate)

Overview

Stub doc generated from the script's in-source header comment. See scripts/run-vm-signing-matrix.sh for canonical behavior. This stub exists so the CM-SCRIPT-DOCS-SYNC pre-push gate (.github/hooks/pre-push Check 9) starts gating modifications to this script.

The script's own header documentation (verbatim):

```
scripts/run-vm-signing-matrix.sh — cross-arch signing matrix wrapper.
```

```
Drives cmd/vm-matrix against (alpine,debian,fedora) × (x86_64,aarch64,riscv64)
```

```
= 9 configs. Each VM signs the same input APK with the same keystore.
```

```
Post-processing computes per-row signing_match by comparing the SHA-256 of /tmp/signed.apk to the alpine-3.20-x86_64 KVM reference.
```

```
Bluff vector this catches: JCA-provider divergence — the same JRE producing different signing bytes across architectures. If signing bytes diverge, an APK signed on architecture A might validate but not match what architecture B produces, breaking reproducible-build guarantees and silently exposing keystore-binding inconsistencies.
```

```
The matrix RUN is the constitutional gate; this wrapper exists so Lava-side CI invocation is uniform (a single bash entry-point) and the post-processing block (the divergence detector) is unit-testable via tests/vm-signing/test_*.sh fixtures.
```

Pre-requisites the operator MUST satisfy before running:

- tests/vm-signing/sample.apk exists (operator supplies; intentionally

NOT shipped – anti-bluff: don't pretend a stub APK is signable)

- proxy/build/libs/app.jar exists (./gradlew :proxy:buildFatJar)
- keystores/upload.keystore.p12 exists (Lava signing keystore)
- tools/lava-containers/vm-images.json has REAL SHA-256 + size

for

each of the 9 qcow2 entries (placeholder zeros reject in pkg/cache)

Exit codes:

- 0 – all 9 rows produced byte-equivalent signed APKs
- 1 – at least one row diverged (JCA bluff caught) OR a row's signing-output.json is missing
- 2 – configuration error (missing inputs, missing tooling)

Usage

See the script's in-source comment block (above) for canonical usage examples.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC requires it). Per §11.4.18, the documentation MUST stay in sync with the codebase — no doc may be out of sync with its script.

Cross-references

- scripts/run-vm-signing-matrix.sh — the script itself
- docs/helix-constitution-gates.md — gate inventory
- HelixConstitution Constitution.md §11.4.18 (the mandate)
- Lava CLAUDE.md §6.AD (HelixConstitution Inheritance)